Garmin can falsify this in one week using existing data.

2026 Garmin Health Awards — Health Care & Research category

A bimodality-detection metric for autonomic sleep stability

3.5 years of personal Garmin monitoring · 4 months post-intervention · software-only detection pipeline · falsifiable population-scale test specified.

Executive summary

The ask. Run a one-week analysis on existing Garmin Connect REM-respiration data (~100 users × 10 nights) to test for structured bimodality using a fully specified k=1 vs k=2 Gaussian mixture model pipeline with false-positive control. The output is binary: either a non-trivial subset of users exhibits stable, above-floor bimodality with within-night structure, or the distribution is indistinguishable from unimodal synthetic data and the hypothesis is rejected. No new data, firmware, sensors, or user action are required. This is a low-cost, decisive test of whether a previously unmeasured autonomic state signal exists at population scale.

Headline findings (n=1)

- **Bimodal-fraction collapse: ~90% → 0%.** Across 46 baseline nights (2022–2025), 86–94% of REM segments show bimodal respiration structure (Ashman's D 2.36–2.40). After intervention: 35% (Mar 2026, partial), then 0% (Apr 2026, full). The most sensitive marker in the dataset and the primary Phase A target.
- **25% reduction in per-night respiratory variability.** In-REM RMS deviation: 1.21 brpm baseline → 0.91 brpm post-intervention (cross-surface average). Sustained for four months.
- **Thermal correlation.** Bimodality severity concentrates in 02:00–05:00, coincident with the nocturnal temperature trough.
- **Stable multi-year baseline.** Three baseline periods spanning 3.5 years (Oct 2022, Mar 2025, Apr 2025) show bimodal fraction 86–94%, mean RMS within 0.01 brpm. The intervention landed on a plateau, not a decline already in progress.

Rubric alignment, in one line each

- **Customer benefit:** an autonomic-recovery readout for Long-COVID (PASC, ~65M globally) and post-TB lung disease cohorts, from data Garmin already collects.
- **Scalability:** software-only, computationally inexpensive, deployable across Garmin's existing device base at zero marginal infrastructure cost.
- **Uniqueness:** a state-transition marker — "which discrete state is the user in" rather than "how much variability" — not captured by any current wearable metric.

- **Innovation:** per-segment GMM bimodality testing with explicit false-positive control, paired with a candidate non-pharmacological modulation pathway.
- **Performance:** bimodal-fraction collapse from ~90% to 0%, sustained four months, on a stable 3.5-year baseline.

The data

Figure 1 — Bimodal to unimodal transition

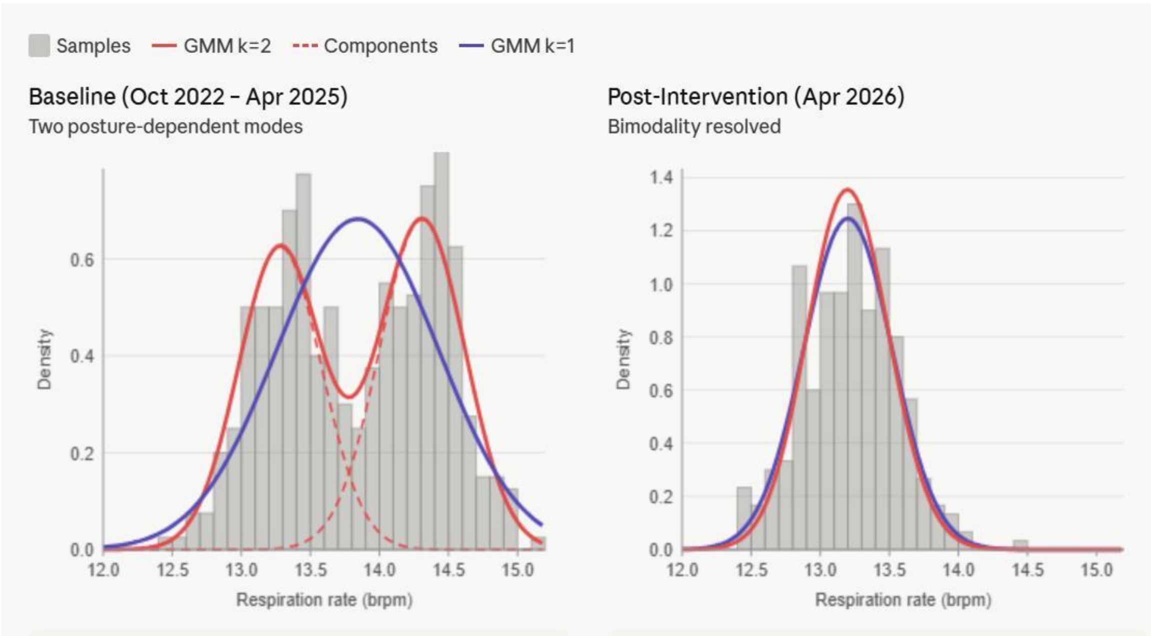


Figure 1. Left panel — baseline distribution shows two clear peaks at ~13.3 and ~14.3 brpm; the red k=2 GMM fit tracks the bimodal structure while the purple k=1 fit visibly fails to capture the two humps. Dashed red lines show each individual Gaussian component. Right panel — post-intervention shows a single clean peak at ~13.2 brpm where k=1 and k=2 fits overlap completely, confirming bimodality is absent.

Figure 2 — In-REM respiratory variability (the supporting signal)

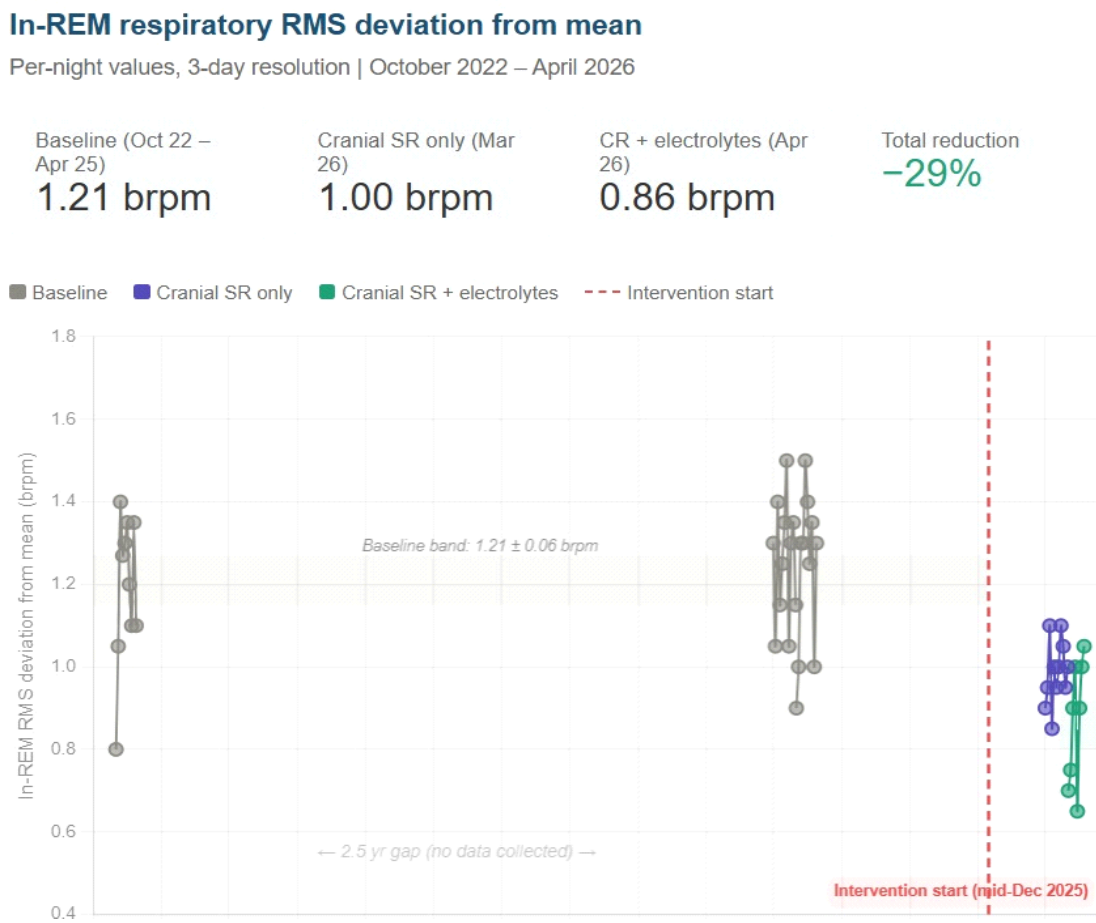


Figure 2. Per-night in-REM RMS deviation from mean. Baseline (Oct 2022 – Apr 2025): 1.21 brpm. Cranial sleep-surface SR only (Mar 2026): 1.00 brpm. Sleep-surface SR + electrolyte logging (Apr 2026): 0.86 brpm. Total reduction ~29% (warm-foam-specific); cross-surface average 25%. Intervention start: mid-December 2025. The Oct 2025 – Jan 2026 gap is a true data outage, not a methodological choice.

The reduction is modest in absolute terms (≈0.30 brpm) but consistent across two intervention phases and stable for four months.

Figure 3 — Within-night temporal structure

Bimodality severity in the n=1 record is not uniform across the sleep period. It concentrates in the 02:00–05:00 window — the coldest hours of a typical Highveld autumn night — and is much weaker in the early-evening and post-dawn windows when ambient temperature is higher. The intervention effect tracks the same time-of-night pattern: the largest absolute reductions occur in the same coldest-hours window in which baseline severity peaks.

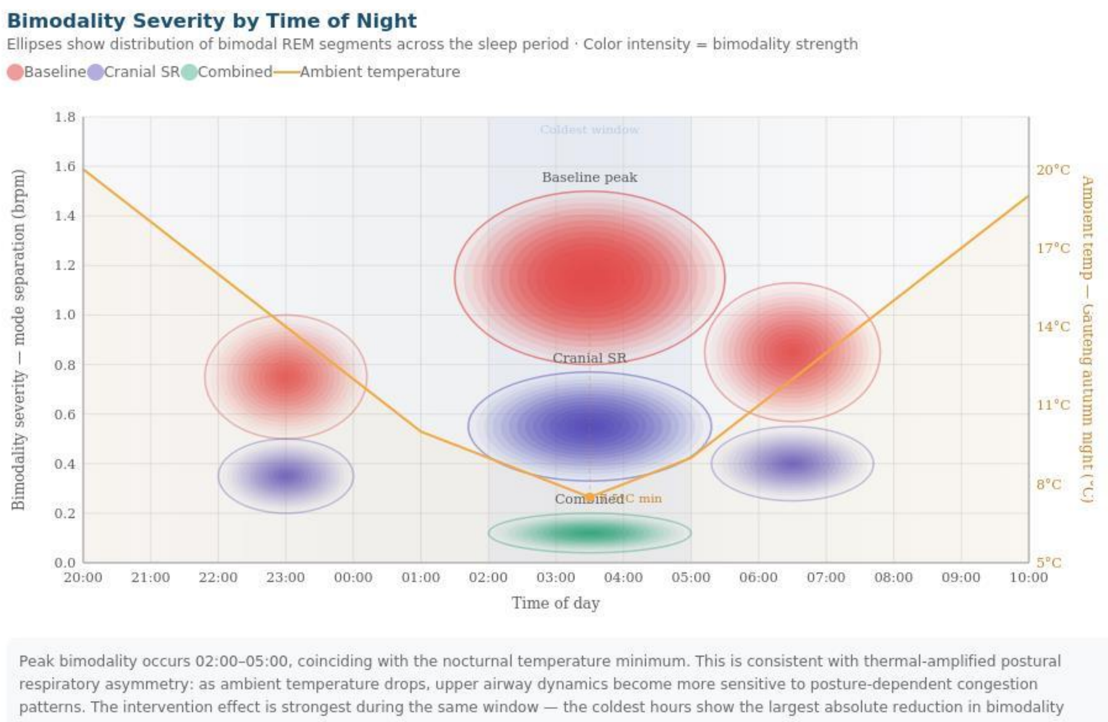


Figure 3. Bimodality severity by time of night, n=1 record. Ellipses show the distribution of bimodal REM segments across the 20:00–10:00 sleep window for three conditions: baseline (red), cranial sleep-surface SR only (purple), and combined intervention (green); ellipse opacity reflects bimodality strength. Orange curve: typical Gauteng autumn overnight ambient temperature profile (right axis). The temperature minimum at ≈04:00 sits directly beneath the baseline bimodality peak.

What this is and is not. The figure shows a within-night temporal correlation between bimodality severity and ambient temperature. It does not, on its own, demonstrate thermal causation — the alignment is consistent with thermal-amplified posture-dependent respiratory asymmetry, but also with circadian-phase effects, sleep-stage architecture (REM concentrates in late-night hours), or ultradian autonomic cycling. The temperature-correlation observation is most usefully read as a falsifiable prediction for Phase A.

Figure 4 — Population context

From 2004 through 2019, public search behaviour for “shortness of breath” followed a stable seasonal pattern in the 12–33 range, decoupled from infection dynamics: even the 2009 H1N1 pandemic produced no detectable shortness-of-breath rise. The 2020 acute COVID-19 pandemic coupled the two terms briefly, with shortness-of-breath spiking 2–3× (peak 100 in

March 2020). Post-pandemic, the coupling has not re-established. Flu has returned to its pre-pandemic seasonal pattern at 4–10. Shortness-of-breath has settled at a sustained 38–55 plateau — approximately double its pre-pandemic baseline — and held that plateau for five years.

In late 2025 and early 2026 this plateau has elevated further. April 2026 reached 83 — approaching pandemic-era peaks during a period of ordinary flu activity and no acute respiratory pandemic. The decoupling, the sustained plateau, and the recent escalation are all consistent with a population reporting respiratory difficulty driven by something other than acute infection.

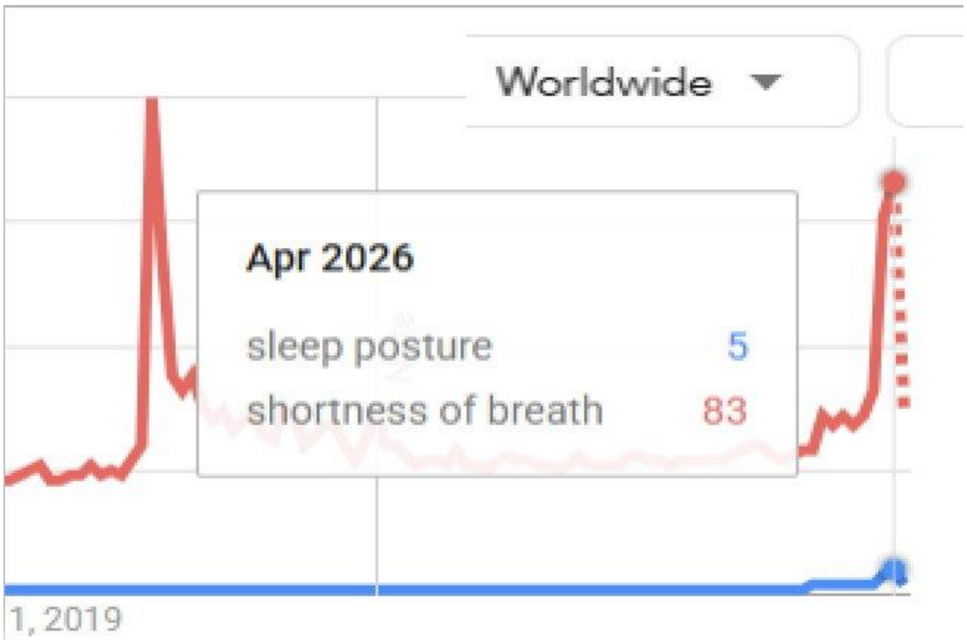


Figure 4. Worldwide Google Trends search activity, September 2019 – May 2026. Red trace: "shortness of breath". Blue trace: "sleep posture". April 2026 readings (popout): shortness of breath 83, sleep posture 5. The acute COVID-19 peak (March 2020, ~100) is visible at the left; the recent acceleration to 83 in April 2026 is not associated with any acute respiratory pandemic.

Calibration note

The 17× ratio between symptom searches (shortness of breath) and candidate causes (sleep posture) is standard calibration, not evidence against postural causation. Because lay search vocabulary for mechanical causes sits below the "noise floor," population search data is an insensitive instrument. Phase A (wearable data) is required because it bypasses language entirely, measuring respiratory structure directly to detect signals that search behaviour cannot. The April 2026 reading at 83% of the COVID-pandemic peak despite no active pandemic confirms that respiratory instability is a broad population-level issue, not a niche interest.

The metric: REM-respiration bimodality

Existing wearable autonomic metrics ask: how much variability does this user have? This submission proposes a different question: which discrete state is this user locked into, and can they be moved?

Unlike aggregated metrics like HRV or RHR, REM-respiration bimodality is an upstream observable. It allows users to decompose autonomic instability into specific, actionable environmental factors — such as bedding surface or mechanical coupling. The primary value is a long-term, per-night metric that surfaces behavioural and environmental contributors to dysregulation, enabling users to make simple, non-technical adjustments to their sleep environment before autonomic instability becomes entrenched. The metric functions as a diagnostic instrument regardless of which specific intervention pathway is utilised.

Metric type	Actionability	User value
Standard (HRV/RHR)	Aggregated / opaque	Difficult to link to specific causes.
Bimodality fraction	Decomposable	Directly links environmental changes to signal shifts.

Methodological transparency

What the metrics are, exactly

- **Bimodal fraction:** per-night fraction of REM segments classified as bimodal under the Phase A test ($\Delta BIC > 6$, both component weights ≥ 0.25, mean separation ≥ 0.5 brpm). Estimated from screenshot analysis in this n=1 record; Phase A would compute it directly from raw samples.
- **In-REM RMS:** per-night in-REM RMS deviation from segment mean, computed from Garmin Connect's exported per-night respiration samples within REM-classified periods. No detrending, no seasonal adjustment, no outlier removal.

What this submission is and is not

This is a single-subject observational record with 3.5 years of pre-intervention baseline and 4 months of post-intervention follow-up, used to motivate a falsifiable population-scale test. It is not a clinical trial, a diagnostic claim, a therapeutic claim, a request for funding or IP discussion, or an assertion of mechanism certainty.

Timeline and provenance

The longitudinal data (2022–2026) accounts for environmental variables, including the El Niño/La Niña cycles and the southern-hemisphere hot season. Three properties make the post-intervention reductions interpretable beyond the raw numbers:

- **Stable baseline.** Multi-year pre-period without trend. The intervention landed on a plateau, not a decline already in progress — ruling out the most common null explanation for n=1 effects.
- **Sustained outcome.** Four months of post-period stability is past the window in which placebo or novelty effects typically decay in autonomic research (≈2–6 weeks).

- **Aircon-free normalisation.** In 2026, the subject maintained stability despite higher ambient temperatures (26 °C vs 22 °C in prior years), suggesting an improvement in intrinsic physiological control rather than environmental cooling.
- **Failed washout (preliminary).** Withdrawal of the SR intervention in April 2026 did not produce an immediate return to baseline variability. This is consistent with a transition into a persistent stable attractor rather than a temporary forcing effect, and is the single most theoretically informative observation in the dataset.

The data gap and its uses

- **Oct 2025 – Jan 2026 outage.** The Garmin device was lent to a family member during this period; the gap is not a methodological choice. The intervention began in mid-December 2025 during the gap, so the resumption of recording in late January 2026 captures the early adaptation phase rather than steady-state response.
- **January 2026 transient overshoot.** On resuming Garmin use after the gap, the data shows a brief period of elevated resting heart rate — an autonomic overcorrection — before settling into the lower-variability regime. This transient is preserved in the data, not removed. It is one reason a firmware artefact is unlikely to explain the figures: a respiration-algorithm change in firmware would produce a step transition, not a transient overshoot followed by gradual stabilisation.

Modulation pathway and disclosure

The candidate intervention pathway used in the n=1 record is mechanical stochastic resonance via a sleep-coupling element, delivered via the residual mechanical signature of a commercially available low-noise fan (MeacoFan family). Prior to public deposition, the author disclosed the secondary-utility observation to both Solenco (the South African distributor for the Meaco range) and to Meaco UK by direct correspondence. Solenco acknowledged the disclosure and indicated it would be forwarded to the Meaco UK Managing Director. The disclosure was made in good faith and without any commercial-terms negotiation; no licensing, partnership, or financial arrangement exists between the author and either party at the time of submission. Detail on coupling signal strength and frequency characterisation is available on request.

Phase A — bimodality detection specification

This is the method the front-matter ask refers to. It is specified at the level of detail needed for a Garmin data scientist to implement directly.

Inputs

- Per-night respiration samples during REM-classified segments only (Garmin's existing sleep staging).
- REM segments of duration ≥10 minutes (primary), with sensitivity analysis at ≥5 minutes.

Per-segment analysis

- Fit Gaussian Mixture Models with k=1 (unimodal) and k=2 (bimodal) to the within-segment respiration distribution.
- Compare via Bayesian Information Criterion (BIC). A segment is classified as bimodal if all of the following hold: $\Delta BIC = BIC(k=1) - BIC(k=2) > 6$; both component weights ≥ 0.25; mean separation between components ≥ 0.5 bpm.

Predicted outcomes (pre-registered)

If the bimodality phenotype exists, Garmin should observe:

- **Non-trivial BI population.** A right-skewed BI distribution with a subset of users consistently above the false-positive floor.
- **Within-user stability.** BI remains relatively stable across nights within the same user (trait-like, not random).
- **Within-night structure.** Elevated bimodality in late-night REM segments (≈02:00–05:00).

If these patterns are not observed, the hypothesis is rejected.

Decision rule (binary, explicit)

- **Positive result:** a measurable subset of users shows BI above false-positive levels with internal structure → proceed to population characterisation (Phase B).
- **Negative result:** BI distribution indistinguishable from synthetic unimodal baseline → hypothesis not supported; no further work warranted.

False-positive control

- Apply the identical pipeline to synthetic unimodal respiration data with matched mean, variance, and segment length distribution. Target false-positive rate: <5%.
- Pre-built ground-truth-labelled validation datasets are referenced in Section 2 of the appendix and are available on request so an engineer can verify their implementation before deploying on real user data.

Per-night and per-user aggregation

- **Bimodality Index (BI).** Fraction of REM segments per night classified as bimodal. Aggregated to a monthly per-user value.
- **Within-user stability.** Standard deviation of BI across nights within a 30-day window — quantifies whether bimodality is a stable user trait or a per-night fluctuation.

Secondary predictions (free with the same pipeline)

Two falsifiable secondary predictions follow from the within-night temporal structure observed in the n=1 record. They cost nothing additional to compute.

- **Time-of-night concentration.** Bimodality should concentrate in late-night REM segments (approximately 02:00–05:00 local time). Stratifying per-segment BI by the segment's timestamp within the sleep period should reveal a peak in this window.
- **Geographic temperature-amplitude correlation.** Across users, BI should correlate with nocturnal temperature amplitude in the user's geography. Garmin already has implicit access to this stratification through user geography.

Phase A scope

≈100 users × 10 nights — enough to characterise false-positive behaviour, BI distribution shape, and visual confirmation of segment-level bimodality. ≈one analyst-week. The output is binary at the level the submission cares about: either the BI distribution is non-trivially populated above the false-positive floor, or it is not. Either result is informative.

Optional zero-cost extension

A secondary, zero-cost extension of the same pipeline is to test whether per-night bimodality indices co-vary with external population-stress signals such as the March–April 2026 shortness-of-breath search-volume acceleration (Figure 4). This does not affect the primary detection result but provides an immediate real-world validation opportunity: if the bimodality phenotype is present and atmospherically modulated, periods of elevated population respiratory-difficulty signal should align with periods of elevated per-user bimodality on the corresponding geographies and dates. The test runs against the same Garmin data being analysed for the primary detection; no additional data collection is required.

Appendix — Section 1: Potential future roadmap

Sketch only — included so a reviewer can see the logical trajectory beyond what the present submission claims. None of this is part of the current entry's scope. Each phase is independently abortable.

Phase B — population characterisation

- Same GMM pipeline applied at scale (≥10^5 users). Outputs: BI distribution, prevalence at the various ΔBIC thresholds, demographic stratification, association with self-reported sleep quality where available.
- Establishes a normative reference for REM bimodality in a large general-population cohort.

Phase C — broadening of phenotype

Once the bimodality phenotype is established at population scale, the next step is to broaden the phenotype by testing whether bimodality co-occurs with adjacent autonomic markers Garmin already records. The n=1 record provides one concrete example: a paired bradycardia / oscillating-respiration pattern that resolved alongside the bimodality collapse.

Worked example: bradycardia resolution in the n=1 record

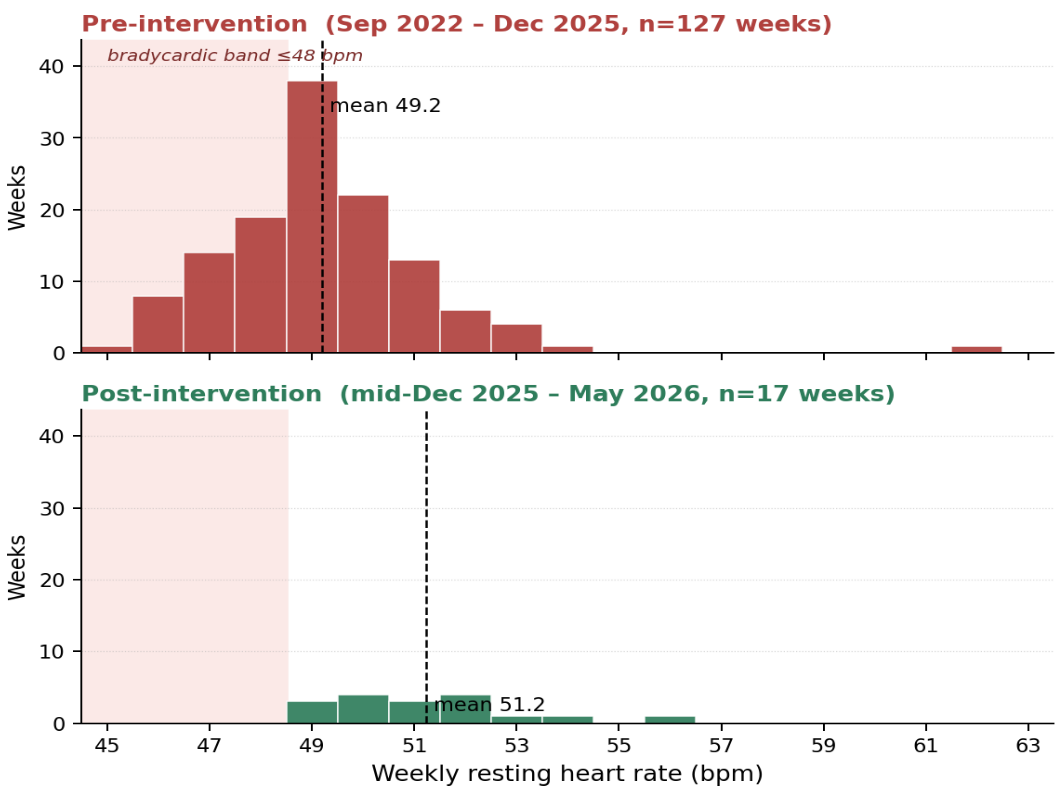


Figure 5. Weekly resting heart rate, stacked pre/post intervention. Top panel (red): pre-intervention weeks, Sep 2022 – mid-Dec 2025 (n=127); mean 49.2 bpm; 33% of weeks fall in the bradycardic band

≤48 bpm (shaded). Bottom panel (green): post-intervention weeks, mid-Dec 2025 – May 2026 (n=17); mean 51.2 bpm; 0% of weeks fall in the bradycardic band. Source: Garmin Connect weekly resting heart rate, automatically computed by the wrist device. Same axes on both panels for direct visual comparison.

A truncated left tail. The pre-intervention distribution shows a substantial left tail extending into the bradycardic band, with weekly RHR reaching 45–47 bpm in 33% of weeks across the 3.25-year baseline. The post-intervention distribution is tighter and right-shifted, with no weekly value below 49 bpm over four months of follow-up. The mean shifts by 2.0 bpm (49.2 → 51.2), but the more interpretable change is the disappearance of the bradycardic shoulder rather than the mean shift itself.

Pairing with the respiratory signal. In the pre-intervention period the bradycardic weekly troughs were repeatedly paired with oscillating breathing-rate traces on the Garmin respiratory graph — a visual pattern consistent with periodic or central-pattern respiratory events during sleep. This pairing is what makes the cardiac signal interpretable rather than incidental: a subject whose resting heart rate occasionally dips into the mid-40s without paired respiratory oscillation is plausibly an athletic resting baseline; the same subject with paired oscillating breathing has a different phenotype. In the post-intervention period the bradycardic excursions disappear and the paired oscillating-breathing pattern has not recurred over four months.

What this illustrates for Phase C. Bimodality, bradycardic excursions, and oscillating breathing-rate traces all moved together in the same direction at the same time in the same subject. A Phase C analysis would stratify the Phase B bimodality population by RHR distribution shape (specifically: presence of a left-tail bradycardic shoulder) and ask whether bimodality status predicts which users carry the paired phenotype. This is broadening, not redefinition: the bimodality metric remains primary; bradycardia and oscillating respiration become candidate co-markers that enrich a phenotypic profile without requiring new sensor data.

Phase D — engaging with the identified phenotype

Once a phenotype is identified at population scale (Phase B) and broadened with co-markers (Phase C), Phase D is about turning that into something the user and the platform can act on. Two steps.

Step 1: HRV – bimodality coexistence, with effect-size example

A reviewer is likely to ask: if the goal is autonomic-state monitoring, why not use HRV, which Garmin already computes? The answer is not to remove HRV — it remains a valuable signal where the hardware supports it — but to position bimodality as a complementary metric occupying a different niche. Three positional reasons:

- **Hardware reach.** HRV monitoring of useful quality requires the more advanced optical and ECG-grade sensors found on a subset of Garmin's product line. REM-respiration sampling is available on a substantially wider range of Garmin devices, including lower-tier models. For a long-term monitoring metric intended to scale across the user base, hardware reach matters more than peak signal quality on the highest-end devices.

- **Closer to the underlying physiology.** HRV is a derivative — a statistical property computed across a window of heart-rate samples. Bimodality of REM respiration is a structural property of the respiration samples themselves, one analytical step closer to the autonomic process being monitored.
- **Decomposable into actionable components.** Bimodality fraction responds visibly and separately to bedding-surface choice, mechanical coupling, and electrolyte logging (see Step 2). A user who changes one variable at a time can see the contribution of each in their own bimodality record. HRV cannot be decomposed in this way.

Within-subject effect-size example: HRV biofeedback (2023–2024) vs SR (2025–2026)

During September 2023 – November 2024 the n=1 subject used real-time HRV biofeedback at sleep onset, with the explicit behavioural goal of maximising HRV in the minutes preceding sleep. The biofeedback signal was Garmin's on-device HRV reading; the behavioural protocol was paced breathing and posture adjustment until the on-device HRV indicator entered its upper band. No mechanical, environmental, or pharmacological intervention was applied; the manipulation was exclusively pre-sleep autonomic self-regulation.

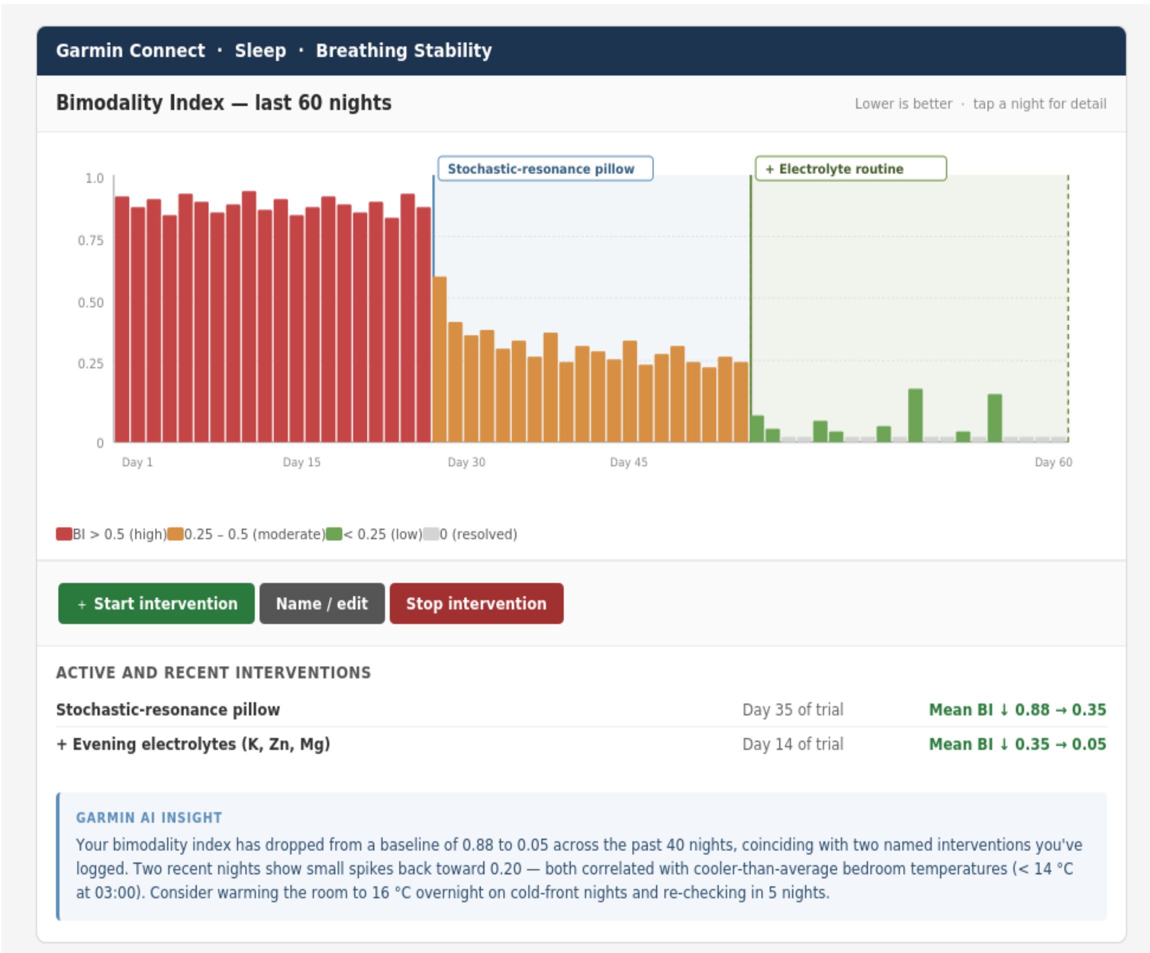
The HRV-biofeedback period produced a slight tightening of the daily REM breathing-rate RMS distribution and a small reduction in its mean compared with the immediately preceding baseline. The effect was visible in the per-night record but modest: per-night bimodality structure was preserved.

Intervention	REM-RMS change	Bimodal-fraction change	Qualitative
HRV pre-sleep biofeedback (Sep 2023 – Nov 2024)	Slight reduction; tighter daily distribution	No detectable change; bimodality preserved	Modest
Mechanical stochastic resonance (Mar – Apr 2026)	1.21 → 0.87 brpm (~25–29% reduction)	~90% → 0%; full collapse	Dominant

Both interventions act on the autonomic-respiratory state, but at different points in the chain. HRV pre-sleep biofeedback is a top-down behavioural manipulation of pre-sleep autonomic tone; the mechanical pathway is a bottom-up environmental manipulation continuously present during sleep itself. In this subject the mechanical pathway dominated effect size on both metrics by roughly an order of magnitude. The two interventions were applied at different times in the same subject, not in a counterbalanced design — this is a within-subject effect-size ranking under naturalistic conditions, not a head-to-head trial. What it does establish is that within this subject, a well-validated mainstream autonomic-modulation technique produced a substantially smaller effect on the same primary metrics than the mechanical pathway. This is why the front-matter ask is for population-scale testing of bimodality detection rather than of the intervention — the detection metric is what generalises; which intervention dominates is a Phase B/C question.

Step 2: Tool-supported intervention tracking — the bedding effect

The bimodality metric only delivers user value if the platform gives users a way to test what affects it. Existing wearable platforms log many physiological metrics but provide essentially no infrastructure for users to test what affects them. A user who changes their bedding configuration, room temperature, evening routine, or supplementation today has no structured way to see whether the change had any measurable effect.



Illustrative Garmin Connect interface. Per-night Bimodality Index over the last 60 nights, with two user-named intervention windows ("Stochastic-resonance pillow" from day 21; "+ Electrolyte routine" from day 41). Three controls — Start intervention, Name / edit, Stop intervention — let the user mark and label any change in their sleep environment or behaviour. Each named intervention accrues a quantified before/after impact. A Garmin AI insight panel (bottom) connects observed bimodality changes to other Garmin-recorded variables (here, bedroom temperature) and suggests low-intrusiveness next actions.

Why this is the landmark step. The proposed interface is the missing piece: a self-service framework in which the user defines the experiment ("Start intervention: cooler bedroom"), the device runs the measurement (per-night Bimodality Index), and the platform synthesises the result ("Mean BI ↓ 0.45 → 0.20 over 14 nights"). Paired with AI-generated explanations that connect bimodality changes to other Garmin-recorded variables, this becomes the blueprint for self-service intervention tracking in personalised health — not specific to bimodality, but enabled by having a metric upstream and decomposable enough to make individual contributions visible.

Why structured tracking matters: the bedding decomposition

To illustrate why structured tool-supported tracking matters, the n=1 record was decomposed into five conditions sampled across the 2022–2026 observation window. Bedding surface (warm foam vs cool spring) is a single variable that produces a substantial bimodality shift on its own — but only structured tracking lets the user see that contribution separately from the mechanical and electrolyte components.

Condition	Surface	SR	Electrolytes	RMS (brpm)	Bimodal %	Reduction vs baseline
3.5y baseline	Spring (cool)	—	—	1.21 ± 0.06	86–94%	—
Oct 2025	Foam (warm)	—	—	1.06	53%	12% RMS
Mar 2026	Spring (cool)	✓	—	1.00	35%	17% RMS
Apr 2026 (a)	Foam (warm)	✓	✓	0.87	0%	28% RMS
Apr 2026 (b)	Spring (cool)	✓	✓	0.94	23%	22% RMS

Reading the decomposition. Surface change alone (Oct 2025) produces approximately half the maximum bimodality reduction without any active intervention: bimodal fraction drops from baseline ~90% to 53%. SR alone (Mar 2026) produces a comparable but slightly larger effect: bimodal fraction 35%. The combination of surface, SR, and electrolytes produces full collapse on warm foam (Apr 2026 a) and near-collapse on cool spring (Apr 2026 b). Surface and SR contribute comparably; neither is the singular driver, and the combination is required for full collapse.

What this means for tool design. Without three-stage intervention tracking, a user who changes both bedding and adds a mechanical-coupling element at the same time cannot know which contributed how much. The five-condition decomposition above was only possible because the n=1 subject had four years of baseline data and could change one variable at a time across multiple months. A user-facing tool with explicit Start / Name / Stop intervention markers — as in the illustrative interface above — lowers the data and time burden by orders of magnitude. This is what makes the bimodality metric actionable rather than merely informative: simple bedding-surface choice is a non-technical, low-cost, immediately user-actionable intervention that produces substantial bimodality reduction on its own, and the tool makes that visible to the user without requiring them to wait years.

Caveat. The two combined-intervention conditions in the decomposition above differ in surface only and in their RMS values by 0.07 brpm and bimodal fraction by 23 percentage points. Three explanations are equally plausible from this data alone: (a) genuine surface-by-intervention interaction favouring warm foam, (b) order effect (foam came first), or (c) within-period measurement noise compounded by screenshot-based scoring. The decomposition is suggestive of a structure that population-scale testing would either confirm or falsify; it is also a worked example of why structured tool-supported tracking would be valuable at scale.

Phase E — deepening modelling: ASSI and extended observer functions

Phases B–D treat bimodality as a single primary metric. Phase E asks what additional structure exists around it — confounder-control instruments, observer functions for instrumented adherence, and the integration path from lab reference into Garmin Connect as a third-party API.

ASSI — Autonomic Sleep Stability Index (confounder-control instrument)

Garmin's data is thermally blind — ambient temperature and humidity are not recorded but are the dominant confounders for autonomic and behavioural sleep metrics on the Highveld. The bimodality finding is thermally robust by construction (within-REM-segment respiration structure does not depend on ambient conditions), which is why it stands on Garmin data alone. The author has a code complete ready to test thermal-domain composite — the Autonomic Sleep Stability Index (ASSI) — that isolates autonomic from thermal effects in the n=1 record using external instrumentation. ASSI exists as a lab-reference instrument first, and as the basis for the integration path described below. Full specification, sub-score definitions, and confidence-weighting protocol are available on request.

Extended observer functions — instrumented adherence platform

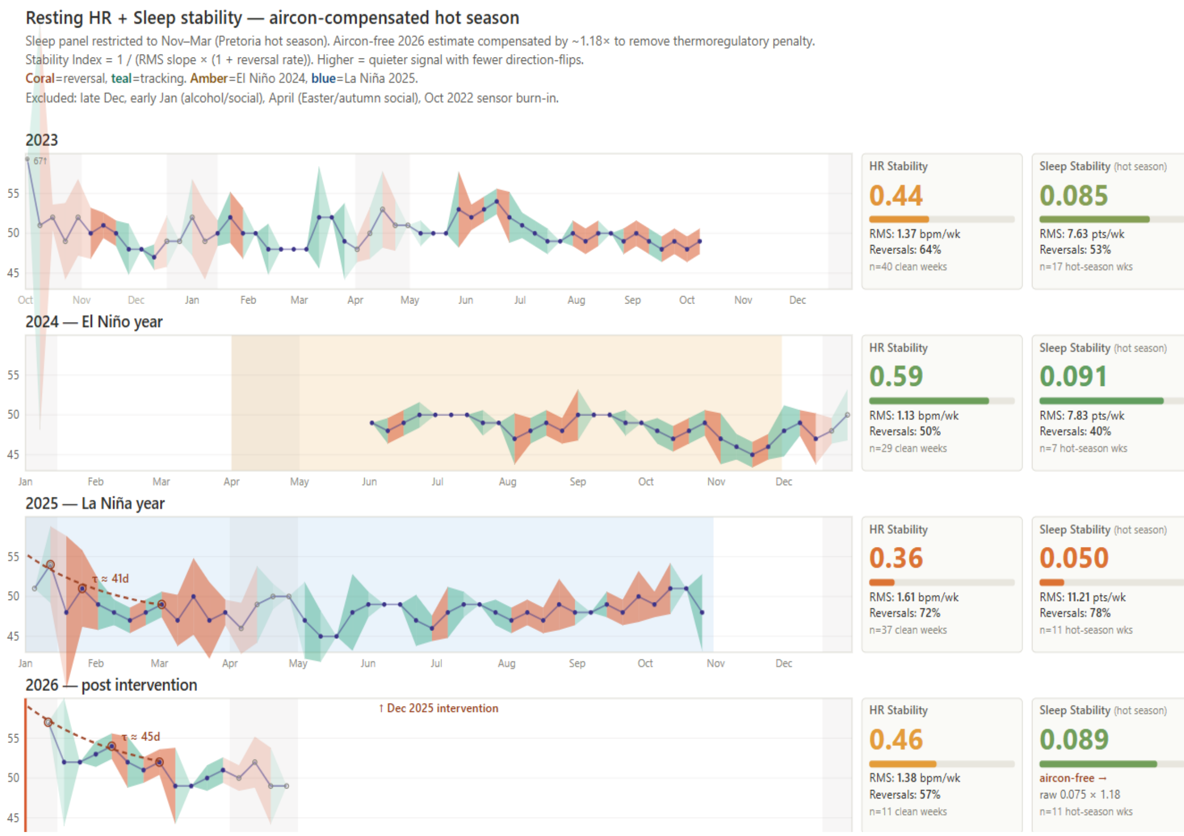
A hardware platform code-ready for testing exists for instrumented adherence monitoring of the candidate intervention pathway: a thermal sensor array (7× TMP117 with SHT41 ambient compensation) and an independent accelerometer-based observer platform (STM32G474 + LSM6DSO IMU) recording surface stiffness, dislodgement events, and intervention duty cycle. Confounder events (late caffeine, transdermal electrolyte logging, illness flags) are timestamped via dedicated buttons. None of this is required to evaluate the front-matter bimodality findings; it exists so a third-party replication conversation has a concrete starting point if Phase A returns positive.

Integration path: lab reference → Garmin Connect third-party API

The natural integration path runs from lab-reference instrument (ASSI + observer hardware) through a dedicated third-party integration into Garmin Connect via the Garmin Health API. In this path, ASSI and the observer functions become external data streams that augment the Garmin-native bimodality metric without competing with it: the device provides the primary signal, the third-party stack provides confounder control and adherence verification, and the user-facing interface (Phase D Step 2) surfaces the combined picture. This is the natural endpoint of the roadmap — a metric originating in Garmin's sensor data, deepened by external instrumentation, and reintegrated into Garmin's user interface as a richer composite.

Provenance and limitations

Provenance — multi-year stability



Provenance figure. Resting heart-rate dynamics and aircon-compensated sleep stability across seasonal and climatic regimes (2023–2026). Weekly RHR (blue) decomposed into directional segments: tracking (teal) and reversal (coral), opacity reflecting magnitude. Climatic regimes shaded: El Niño 2024 (amber), La Niña 2025 (blue). The 2026 panel shows the post-intervention condition holding despite no air conditioning.

Stable baseline confirmation. Across 2023–2025 baseline years (covering both El Niño and La Niña regimes) the resting-heart-rate signal shows variability without monotonic trend; the intervention landed on a plateau, not a baseline already in motion. The 2026 post-intervention condition holds despite the additional thermal stress of no air conditioning. Full year-by-year stability indices, exponential time-constant fits to recovery segments, and the aircon-normalisation methodology are available on request.

AI use during compilation of this submission

AI was used intensively to convert screen shots to the different view presented in this document and in the drafting of the document itself.

That leads to the question in what way is AI used during the judging process.

Here what I observed with AI:

It reduces its thinking space to.fall within existing organizational. Structures and manmade rules.

It does not include what ifs of human sovereignty over existing structures and rules. Hence it provides a compressed form of a future reality which favours current structures and rules.

With many people not realizing this AI strengthens the lock in into current paradigms suppressing fundamental societal advancement.

The table below shows two columns. The first one following current paradigms the second one equilizing the asymmetry of n=1 submissions by doing the phase A as part of the competition judgement to speed up potential "population level benefit" prospects. (this AI subsection was not subjected to the AI prospects rating and was added manually afterwards).

Prospects of being invited to Bangkok finals

	Current Garmin health competition submission (n-=1)	Because of the mapping onto recent population "shortness of breath" worldwide surge the judges may make the call and run the phase A even before their judging and if they then find the signal	
Claude	14%	82%	82,93%
Grok	17%	67%	74,63%
Chatgpt	27%	80%	66,25%
Deep seek	35%	65%	46,15%
Gemini	65%	92%	29,35%
Average	32%	77%	

Claude acknowledged the performing the phase A as part of the judgement process was not part of its logic paths it pursued. It bumped its base prospects from 9% to 14% as part of acknowledging it as a plausible path.

Limitations

- **Single subject.** A single intervention date, a single device. No cross-subject replication, no blinding, no completed washout within this dataset. This is precisely why a population-scale binary test is proposed rather than a stronger claim.
- **No formal confidence intervals on the headline numbers.** Segmented regression and bootstrap CIs are available on request but not reported on the headline pages because they add statistical theatre to what is fundamentally a visual step-change.

- **Subject–experimenter overlap.** The author designed the instrument, ran the intervention, processed the data, and wrote the interpretation. The author also has personal long-COVID symptoms originating in February 2020. Third-party Phase A replication is the primary cure.
- **Data gap.** The Oct 2025 – Jan 2026 outage means the intervention onset itself is not directly captured in the data; only its early-adaptation phase and steady-state response.

What would change the picture

Up.

- Phase A finds the same bimodality signature in a non-trivial fraction of users with comparable autonomic-recovery profiles.
- A formal washout (≥6 weeks off) produces the predicted rise in variability, followed by a fall on restoration.
- A second subject using the replication instrument reproduces both the variability step and a matching thermal-domain signature.

Down.

- The same variability step appears in users with no reported intervention (i.e. generic to long observation windows).
- Phase A returns no above-floor BI population.
- Formal washout fails to produce the predicted variability rise on extended absence.

Honest summary

A carefully observed single-subject record produces two figures that are unusually clean for the design class, with a coherent leading/lagging relationship between them. It is not proof — it is a candidate for a testable generalisation. If Phase A finds no signal, the result remains a personal observation. If positive, it is a starting point, not an endpoint.

Appendix — Section 2: Detailed dataset and synthetic equivalents

Two resources for Garmin engineers implementing the Phase A bimodality detection pipeline: (1) empirically estimated GMM parameters and per-period detail derived from screenshot analysis across the four real measurement periods, and (2) synthetic validation datasets generated from those parameters for algorithm sandbox testing.

S2.1 — Per-period detail: real datasets

Parameters derived by visual assessment of the Garmin Connect sleep-timeline respiration trace within REM-classified segments. Each night was scored for: REM mean respiration rate, RMS deviation from REM mean, presence/absence of bimodal structure, estimated mode separation when bimodal, and estimated within-mode variability. Period-aggregate values shown.

Period	Condition	Nights	Mean RMS (brpm)	Bimodal fraction	Mean separation (brpm)	Within-mode σ (brpm)
Oct 2022	Baseline	15	1.20	93%	1.02	0.31
Mar 2025	Baseline	17	1.21	94%	1.03	0.31
Apr 2025	Baseline	14	1.21	86%	1.07	0.31
Mar 2026	Cranial SR only	17	1.00	35%	0.73	0.35
Apr 2026	CR + electrolytes	11	0.87	0%	—	—

Key observations across the real datasets

Baseline stability across 3.5 years. The three baseline periods (Oct 2022, Mar 2025, Apr 2025) show remarkable consistency: mean RMS within 0.01 brpm (1.20–1.21), bimodal fraction between 86–94%, mean mode separation between 1.02–1.07 brpm, and within-mode sigma stable at 0.31 brpm. This consistency spans different seasons, different years, and different screenshot formats.

The bimodal-fraction collapse is the headline. The transition from ~90% bimodal nights (baseline) to 35% (cranial SR only) to 0% (combined intervention) is the most dramatic shift in the dataset. The bimodal fraction is the more sensitive metric for population-scale detection and is the primary target for Phase A.

Ashman's D values. D = 2.36–2.40 during baseline indicates well-separated modes under standard statistical criteria (D > 2 is conventionally considered separable). During cranial SR only, D drops to 1.47 — still detectable but marginal. During the combined intervention, bimodality is absent entirely.

Within-mode sigma is the critical implementation parameter. At 0.31 brpm, within-mode variability is low enough that a 1.0 brpm separation produces clearly distinct modes. The signal-to-noise ratio within REM segments is favourable for bimodality detection even at Garmin's typical 30-second sampling resolution.

S2.2 — Expected GMM parameters for Phase A implementation

For a bimodal user (baseline phenotype):

- Component 1 mean: REM_mean − 0.5 brpm
- Component 2 mean: REM_mean + 0.5 brpm
- Component σ each: ~0.30 brpm
- Component weights: approximately equal (0.45–0.55 each)
- Expected ΔBIC: >6 for segments ≥10 min with ≥20 samples
- Expected Ashman's D: 2.0–2.5

For a unimodal user (resolved or healthy):

- Single component mean: REM_mean
- Single component σ: ~0.30–0.40 brpm
- Expected ΔBIC: <2 (k=1 preferred)

S2.3 — Synthetic validation datasets (Phase A sandbox)

Four synthetic datasets are available on request, each containing multiple nights of simulated REM respiration data with ground-truth labels for every REM segment, enabling direct computation of sensitivity and specificity. Bimodal segments use two Gaussian components with means at (base_rate ± separation/2), each with σ = within_sigma; posture switching is modelled as a Markov process with per-sample switching probability of 0.015–0.025. Sampling at 30-second intervals matches Garmin's typical respiration sampling rate.

Each synthetic dataset is parameterised to match one of the four real-period phenotypes characterised in S2.1, plus a unimodal control. The mapping is:

Synthetic dataset	Mirrors real period	Bimodal fraction (target)	Component separation (target)	Within-mode σ (target)
S-Baseline	Oct 2022 / Mar 2025 / Apr 2025 (pooled)	~90%	1.0 brpm	0.31 brpm
S-Partial	Mar 2026 (cranial SR only)	~35%	0.73 brpm	0.35 brpm
S-Resolved	Apr 2026 (combined intervention)	0%	—	0.30–0.40 brpm
S-Unimodal control	n/a (false-positive floor test)	0% (by construction)	—	0.31 brpm

The S-Unimodal control is the calibration anchor: applying the Phase A pipeline to S-Unimodal returns the false-positive rate, which the pipeline must hold below 5%. The other three synthetic datasets test sensitivity at three points along the bimodality intensity spectrum (full, partial, resolved). An engineer can verify their Phase A implementation by running it against all four synthetic datasets and confirming: <5% false positive on S-Unimodal; high recall on S-Baseline; partial detection on S-Partial; near-zero on S-Resolved.

About the author and prior work

The author is an electronic product design engineer based in Pretoria, South Africa. Prior research has focused on physiological adaptation to extreme-climate conditions on the Highveld, including a regional health-tech programme finalist position (The Innovation Hub, Pretoria) for a thermoregulation product. The respiration biomarker findings reported here emerged from a multi-year personal research programme on autonomic recovery in extreme-climate conditions, in the context of long-COVID symptoms persisting since February 2020.

The longitudinal three-and-a-half-year Garmin record that underlies this submission was therefore not collected for the bimodality analysis; it was collected as part of a longer-running interest in autonomic-environmental interactions, and the bimodality finding emerged from re-analysis of an existing dataset. This sequencing is methodologically helpful: the data was not collected with the present hypothesis in mind, which reduces the risk of subtle measurement biases that arise when a researcher knows in advance what they expect to find.